

PSY405 - Lesson 35

The Stimulus–Response Theory of Dollard and Miller

Dollard and Miller tried to **combine behaviorism (learning through habits)** with **psychoanalysis (conflict, anxiety, unconscious)**.

👉 Their main idea is simple but powerful:

| Personality = a collection of learned habits (Stimulus → Response links)

Everything you feel, think, fear, or do is learned through:

- **Drives** (needs)
 - **Cues** (signals)
 - **Responses** (actions or thoughts)
 - **Rewards** (relief or satisfaction)
-



KEY CONCEPT: HABIT (MOST IMPORTANT)

◆ What is a Habit?

Habit = a stable connection between a stimulus (cue) and a response

📌 Scientific definition:

A habit is a learned association between a stimulus and a response.

📌 Plain English:

When **something happens**, you **automatically react in a certain way** because you learned it earlier.

Examples:

- Phone notification → you check your phone
- Exam announced → anxiety
- Parent's angry voice → fear

⚠ Important:

Habits are **not only external** (things outside you), they can also be **internal**:

- Thought → emotion
 - Feeling → behavior
-



CORE STRUCTURE OF THE THEORY

Dollard & Miller explain personality through **four main areas**:

1. **Structure of Personality**
2. **Dynamics of Personality**
3. **Development of Personality**
4. **Applications (therapy, conflict, unconscious)**

We'll focus deeply on **Development of Personality**, because that's the heart of this lesson.



3. DEVELOPMENT OF PERSONALITY

(How a simple baby becomes a complex adult)

They say:

| Personality develops through learning, not mysterious instincts.

I. INNATE EQUIPMENT (What the baby is born with)

◆ Meaning of "Innate"

Innate = **inborn**, present at birth.

The baby is not empty—but also not complex.

1 Reflexes (Automatic responses)

 Definition:

Reflex = automatic response to a specific stimulus.

Examples:

- Dust in nose → sneeze
- Cold → shiver
- Dust in eye → tears

 This is pure **Stimulus → Response**

2 Innate Hierarchies of Responses

 Meaning:

Some responses are **more likely** than others in certain situations.

Examples:

- Loud noise → cry
- Hunger → cry
- Wet diaper → cry

This means:

| The baby has a preferred response order

This is called:

 **Initial hierarchy of responses**

3 Primary Drives

 Definition:

Primary drives = biological needs linked to survival.

Examples:

- Hunger
- Thirst
- Sleep
- Pain avoidance

These drives **motivate behavior**.

Extinction of Old Associations

 Extinction = weakening or removal of an old habit.

Example:

- Baby stops crying at every small noise because it learns it's harmless.



II. THE LEARNING PROCESS

What is Learning (According to Dollard & Miller)?

 Scientific definition:

Learning is the process by which a **stimulus and response become connected**.

 Simple words:

You learn when **something leads to a reaction that brings relief or reward**.



FOUR STEPS OF LEARNING (VERY IMPORTANT)

Learning only happens when **all four steps occur**:

Step	Scientific Term	Simple Meaning
1	Drive	Want or need
2	Cue (Stimulus)	Notice something
3	Response	Do something
4	Reward	Get relief/satisfaction

Example (Studying for Exam)

- 1 Drive → Want to pass
 - 2 Cue → Exam date announced
 - 3 Response → Study
 - 4 Reward → Good grade / relief
- ➡ Habit formed:

Exam → Study

Response Hierarchies

Initial Hierarchy

Responses most likely to occur **before learning**

Example:

- Exam stress → panic, avoid studying

Resultant Hierarchy

Responses **after learning**

Example:

- Exam stress → make timetable, study

 Learning **reorders** responses.

III. SECONDARY DRIVES

What is a Secondary Drive?

 Definition:

A **learned drive** that develops from primary drives.

 Simple words:

Something that was neutral becomes **emotionally powerful**.

Example: Fear & Phobia

- Shock → fear (primary response)
- Room where shock occurred → fear
- Later, even thinking about the room → fear

➡ The room becomes a **secondary drive**

⚠ Secondary drives act **just like primary ones**.



IV. HIGHER MENTAL PROCESSES

Now behavior becomes **mental**, not just physical.

Two Types of Responses

1 Instrumental Responses

👉 Direct actions that affect the environment

Example: Running away, studying, shouting

2 Cue-Producing Responses

👉 Thoughts, labels, words that guide behavior

📌 MOST IMPORTANT: **Language**

📌 Labeling

Labeling = naming situations.

Example:

- Exam = "threat"
- America = "danger"

Once labeled:

➡ Same emotional response appears in different situations

Role of Culture & Society

Language is learned socially.

Society teaches:

- How to think
 - How to label experiences
 - How to solve problems
-



V. CRITICAL STAGES OF DEVELOPMENT

Dollard & Miller agree with Freud here:

| Early childhood experiences shape adult personality.

Especially:

- Conflicts
- Anxiety
- Fear

These become **unconscious habits**.



VI. SOCIAL CONTEXT

Parents unintentionally teach **neurotic conflicts**.

Why?

Because society gives **contradictory demands**:

- Be aggressive ✗
- Be polite ✓
- Be independent ✗
- Obey elders ✓

Child gets confused → frustration → anxiety

⚠ Child lacks language to understand → emotional problems

Same thing can happen in adults during:

- War
- Trauma
- Extreme stress

4. APPLICATIONS OF THE MODEL

i. UNCONSCIOUS PROCESSES

Dollard & Miller accept:

✓ Unconscious behavior exists

✗ But it is **learned**, not instinctual

Unconscious = habits learned early and no longer verbalized.

ii. CONFLICT

Conflict = **two incompatible tendencies at the same time**

▲ Gradients (IMPORTANT EXAM AREA)

Term	Meaning
Gradient of approach	Desire increases as you get closer
Gradient of avoidance	Fear increases as you get closer

⚠ Avoidance is **stronger** than approach.

Types of Conflict

1 Approach–Avoidance Conflict

Same object → attraction + fear

Example:

- Love someone ❤️
- Afraid of rejection 😞

➡ Result: Anxiety

2 Avoidance–Avoidance Conflict

Two bad choices

Example:

- Climb → fear of height
- Don't climb → fear of being called coward

➡ Result: Vacillation (moving back and forth)

😞 HOW NEUROSIS IS LEARNED

Neurosis = learned anxiety patterns

Parents mishandle:

- Feeding
- Toilet training
- Sex education
- Aggression control

Child learns:

- Guilt
- Fear
- Anxiety

➡ Continues into adulthood

5. PSYCHOTHERAPY

Goal:

 **Unlearn maladaptive habits**

Therapist's Role

- Non-punitive
- Sympathetic
- Encourages free expression
- Helps patient understand how habits formed

👉 Therapy = relearning S-R connections



6. RESEARCH

- Strong laboratory support
 - Human & animal experiments
 - Focus on stimulus generalization & displacement
-



7. SUMMARY

- Personality = learned habits
 - Habit = stimulus-response link
 - Learning needs drive, cue, response, reward
 - Conflicts and neurosis are learned
 - Therapy = relearning
-



8. EVALUATION

Strengths

✓ Scientific

- ✓ Logical
- ✓ Experimentally supported

Weakness

- ✗ Too laboratory-based
 - ✗ Ignores creativity and emotions beyond learning
-

Short-Question Answers

Q1: What is the key concept of Dollard and Miller's theory?

Answer:

The key concept of Dollard and Miller's theory is the **habit**, which represents a stable connection between a stimulus (cue) and a response. A habit is a learned association in which the presence of a particular stimulus automatically triggers a specific response. Habits can form between external stimuli and overt (observable) responses, as well as between internal stimuli and responses such as thoughts and emotions.

Explanation (Plain English):

Think of a habit as a **mental shortcut**. When something happens (stimulus), your mind or body reacts in a way you have learned before (response). These can be actions you do, or feelings you have.

Example:

- Hearing your phone ring (stimulus) → checking your phone (response)
- Thinking about exams (stimulus) → feeling anxious (response)

Tips:

- Memorize: "**Stimulus → Response = Habit**"
 - Habits are learned; not born.
-

Q2: What is the role of innate equipment in personality development?

Answer:

Innate equipment refers to the **inborn reflexes, primary drives, and response tendencies** that a baby possesses at birth. These include reflexive actions (like sneezing), hierarchies of responses (some responses are more likely in certain situations), and primary drives such as hunger, thirst, or sleep. Innate equipment provides the foundation for learning and the formation of habits.

Explanation (Plain English):

Babies are not empty slates. They are born with automatic responses and basic needs. These inborn responses guide how they learn new behaviors as they grow.

Example:

- Loud noise → baby cries (innate response)
- Hunger → baby cries or seeks food (primary drive)

Tips:

- Remember: **"Innate equipment = foundation for all learning"**
- Think: reflexes + drives = starting point for habits.

Q3: What is the four-step learning process according to Dollard and Miller?

Answer:

The four-step learning process involves:

1. **Drive** – a motivation or need that energizes behavior.
2. **Cue (Stimulus)** – the signal or event that indicates the opportunity to respond.
3. **Response** – the action or behavior performed to satisfy the drive.
4. **Reward** – the outcome that satisfies the drive and reinforces the habit.

Learning occurs when all four steps are present, creating a stable stimulus-response (S-R) association.

Explanation (Plain English):

To learn something, you must **want it, notice it, act on it, and get satisfaction from it**. This is how habits are formed.

Example:

- Hunger (drive) → sees food (cue) → eats (response) → feels full (reward)

Tips:

- Memorize the order: **Drive → Cue → Response → Reward**
 - You can think of it like **"Motivation → Signal → Action → Satisfaction"**
-

Q4: What are secondary drives, and how do they differ from primary drives?

Answer:

Secondary drives are **learned drives** that develop from experiences and are associated with cues that were originally neutral. Primary drives are biological and innate, like hunger or thirst, whereas secondary drives are acquired through learning and often involve social or emotional motivation.

Explanation (Plain English):

A secondary drive is something you **learn to want or fear** because it becomes associated with your primary needs.

Example:

- Primary drive: fear from a shock
- Secondary drive: fear of a room where the shock occurred

Tips:

- Primary = born with it
 - Secondary = learned from experience
-

Q5: What is the difference between instrumental and cue-producing responses?

Answer:

- **Instrumental responses** are actions that directly affect the environment.

- **Cue-producing responses** are internal or mental responses that help guide future actions by producing stimuli (signals) that lead to another response.

Explanation (Plain English):

Some responses **change the world**, while others **change your thinking or feelings**, which then guide future actions.

Example:

- Instrumental: Running away from danger
- Cue-producing: Naming a situation as "dangerous" → guides later behavior

Tips:

- Think: **Instrumental = Do**
- Cue-producing = **Think/Label**

Q6: How does conflict develop according to Dollard and Miller?

Answer:

Conflict arises when two incompatible tendencies or drives compete simultaneously. Dollard and Miller describe conflict in terms of **approach and avoidance gradients**:

1. **Gradient of approach:** Desire to achieve a goal increases as one gets closer to it.
2. **Gradient of avoidance:** Fear of a negative outcome increases as one gets closer to it.
3. Avoidance is usually **stronger than approach**, causing hesitation or anxiety.

Explanation (Plain English):

Conflict happens when you **want something but also fear it**. The closer you get, the stronger both feelings become, making decisions difficult.

Example:

- A man is attracted to a woman (approach) but feels embarrassed in her presence (avoidance).

Tips:

- Remember: **Avoidance > Approach**
 - Types: Approach–Avoidance, Avoidance–Avoidance
-

Q7: How are neuroses learned according to this theory?**Answer:**

Neuroses are learned during childhood through **improper handling of critical learning situations** by parents. These include:

- Feeding
- Toilet training
- Sex education
- Controlling aggression

Mistakes in these areas lead to **anxiety, guilt, or fear**, which persist into adulthood as neurotic conflicts.

Explanation (Plain English):

Children don't just inherit neuroses; they **learn anxiety and fear** from the way parents respond to their needs and mistakes.

Example:

- A child punished for natural aggression may learn guilt and fear of expressing anger, leading to adult anxiety.

Tips:

- Critical stages = first 6 years
 - Link learning mistakes → adult neurosis
-

Q8: What is the role of psychotherapy in Dollard and Miller's theory?**Answer:**

Psychotherapy focuses on **unlearning maladaptive stimulus-response connections**. The therapist acts as a **sympathetic listener**, helping the patient:

- Express feelings
- Understand how habits formed
- Replace old maladaptive habits with adaptive ones

Explanation (Plain English):

Therapy is like **retraining your brain**. You figure out which habits are causing problems and learn new, healthier responses.

Example:

- Anxiety about exams → explore past learned fear → practice calm studying techniques

Tips:

- Therapy = relearning habits
 - Therapist must be **non-judgmental and supportive**
-

Q9: What is the initial hierarchy of responses and resultant hierarchy of responses?

Answer:

- **Initial hierarchy:** The order of likelihood of responses before learning occurs (innate or natural tendencies).
- **Resultant hierarchy:** The order of responses after learning and experience, reflecting which responses are now stronger or preferred.

Explanation (Plain English):

Before learning, you naturally react a certain way. After learning, your reactions change because some responses are reinforced more than others.

Example:

- Initial: Baby cries at every noise
- Resultant: Baby ignores harmless noise but cries at dangerous noise

Tips:

- Initial = "born tendencies"
 - Resultant = "learned tendencies"
-

Q10: What are the strengths and limitations of the stimulus-response theory?**Answer:****Strengths:**

- Scientific and testable
- Clear link to natural science
- Explains behavior through learning and habit formation

Limitations:

- Focused on lab experiments; less real-life observation
- May oversimplify human behavior and creativity

Explanation (Plain English):

This theory is **powerful and logical**, but it doesn't fully capture emotions, creativity, or messy real-life experiences.

Example:

- Strength: Can study fear conditioning in lab
- Limitation: Doesn't fully explain artistic talent

Tips:

- Strength = scientific rigor
 - Limitation = real-life complexity
-

1. True/False MCQs

Q1: According to Dollard and Miller, a habit is only formed between external stimuli and observable responses.

- A) True
- B) False

Answer: B) False

Explanation: Habits can also form between internal stimuli (like thoughts or emotions) and responses. It's not limited to external behaviors.

Q2: Secondary drives are innate and present at birth.

- A) True
- B) False

Answer: B) False

Explanation: Secondary drives are **learned** from experiences, unlike primary drives which are innate.

Q3: The initial hierarchy of responses refers to learned behavior.

- A) True
- B) False

Answer: B) False

Explanation: The initial hierarchy is **innate tendencies** or natural probability of responses **before learning**.

Q4: Avoidance tendencies are usually stronger than approach tendencies in conflict situations.

- A) True
- B) False

Answer: A) True

Explanation: Dollard and Miller noted that avoidance gradients are steeper, meaning fear/avoidance tends to dominate over approach.

2. Conceptual MCQs

Q5: Which of the following is NOT a component of the four-step learning process?

- A) Drive
- B) Cue
- C) Punishment
- D) Reward

Answer: C) Punishment

Explanation: The four steps are **Drive → Cue → Response → Reward**. Punishment is not a required step in habit formation according to this theory.

Q6: Cue-producing responses are primarily associated with:

- A) Direct action on the environment
- B) Guiding future responses internally
- C) Eliminating primary drives
- D) Reflexive behavior

Answer: B) Guiding future responses internally

Explanation: Cue-producing responses generate signals (like labeling or thoughts) that help guide subsequent actions.

Q7: Which of the following best defines a secondary drive?

- A) Innate biological need
- B) Learned motivation associated with previously neutral stimuli
- C) Reflexive response
- D) Extinction of a habit

Answer: B) Learned motivation associated with previously neutral stimuli

Explanation: Secondary drives arise from learning and can motivate new behavior just like primary drives.

Q8: In Dollard and Miller's theory, which of the following is considered the **primary source of neurosis**?

- A) Genetic inheritance

- B) Social learning during critical childhood stages
- C) Spontaneous internal conflict
- D) Adult trauma only

Answer: B) Social learning during critical childhood stages

Explanation: Neurotic conflicts are learned, mostly during infancy and early childhood, from parental and social interactions.

3. Situational MCQs

Q9: A child is afraid of dogs because he was bitten once. Now, even seeing a picture of a dog makes him fearful. This is an example of:

- A) Primary drive
- B) Secondary drive
- C) Instrumental response
- D) Cue extinction

Answer: B) Secondary drive

Explanation: The fear of dogs (originally neutral) has become a learned motivation due to previous experience, making it a secondary drive.

Q10: A student wants to study for exams (drive), sees the exam date (cue), studies (response), and gets good grades (reward). This demonstrates:

- A) Conflict theory
- B) Four-step learning process
- C) Primary reflex
- D) Cue extinction

Answer: B) Four-step learning process

Explanation: This example shows the **complete S-R learning cycle**: motivation, cue, response, and reward.

Q11: A man is attracted to a woman but feels embarrassed and anxious around her. According to Dollard & Miller, this is an example of:

- A) Avoidance-Avoidance conflict
- B) Approach-Approach conflict
- C) Approach-Avoidance conflict
- D) Gradient extinction

Answer: C) Approach-Avoidance conflict

Explanation: He wants to approach (attraction) but also wants to avoid (fear/embarrassment).

Q12: During therapy, a psychologist helps a patient identify habits learned in childhood and develop healthier responses. Which principle of Dollard & Miller is being applied?

- A) Secondary drive formation
- B) Response hierarchy creation
- C) Psychotherapy as relearning S-R connections
- D) Gradient of avoidance

Answer: C) Psychotherapy as relearning S-R connections

Explanation: Therapy helps **unlearn maladaptive habits** and replace them with adaptive responses.

4. Tricky / Conceptual MCQs

Q13: Which of the following statements correctly describes the relationship between primary and secondary drives?

- A) Secondary drives replace primary drives entirely.
- B) Secondary drives develop through experience and function like primary drives.
- C) Primary drives are irrelevant after early childhood.
- D) Secondary drives are genetically inherited.

Answer: B) Secondary drives develop through experience and function like primary drives

Explanation: Secondary drives are **learned motivations**, often social or emotional, that act similarly to primary drives.

Q14: Which of the following best explains the “resultant hierarchy of responses”?

- A) Innate responses before any learning
- B) Learned and reinforced responses after experience
- C) Unconscious reflexes
- D) Random behavior

Answer: B) Learned and reinforced responses after experience

Explanation: After learning, responses are ranked according to their probability, forming the **resultant hierarchy**.

Q15: A child is punished for expressing anger at home but rewarded for suppressing it. In adulthood, the person feels guilt when asserting themselves. This best demonstrates:

- A) Instrumental response
- B) Learned neurosis
- C) Primary drive conflict
- D) Approach-Approach conflict

Answer: B) Learned neurosis

Explanation: Early mismanaged parental training in critical situations leads to **anxiety and guilt**, forming neurotic habits.

CHEAT SHEET

1 Key Terms & Definitions

Term	Definition	Example
Habit	Stable link between a stimulus (cue) and a response	Hearing phone → checking it
Stimulus (Cue)	Signal or event that triggers a response	Exam date announced
Response	Action or reaction to a stimulus	Studying for exams
Drive	Motivation or need that energizes behavior	Hunger, desire to succeed
Primary Drive	Innate biological needs	Hunger, thirst, sleep
Secondary Drive	Learned motivation from experience	Fear of dogs after being bitten
Initial Hierarchy of Responses	Innate order of likely responses before learning	Baby cries at loud noise
Resultant Hierarchy of Responses	Learned order of responses after experience	Baby cries only at dangerous noise
Instrumental Response	Action affecting environment directly	Running, shouting
Cue-Producing Response	Internal/mental response guiding future actions	Labeling situation as "dangerous"
Neurosis	Learned anxiety or conflict from early experiences	Guilt when expressing anger
Gradient of Approach	Desire increases as goal is approached	Motivation rises as exam nears
Gradient of Avoidance	Fear increases as negative stimulus is approached	Fear grows near a barking dog

2 Four-Step Learning Process (S-R Learning)

Mnemonic: D-C-R-R → Drive, Cue, Response, Reward

Step	Description	Example
Drive	Motivates action	Hunger → want food
Cue (Stimulus)	Triggers response	Smell of food
Response	Behavior/action	Eat food

Step	Description	Example
Reward	Satisfaction/reinforcement	Full stomach

💡 **Tip:** Learning occurs only when **all four steps** are present.

3 Primary vs Secondary Drives

Feature	Primary Drive	Secondary Drive
Origin	Innate	Learned
Example	Hunger, sleep	Fear of dogs, desire for money
Function	Survival	Motivates new learning like primary drives

Memory Hook:

Primary = Born → Basic Needs

Secondary = Learned → Life Skills & Experiences

4 Response Hierarchies

Type	Definition	Example
Initial	Likely responses before learning	Baby cries at every noise
Resultant	Learned responses after experience	Baby cries only at real danger

Mnemonic: I → Initial = Innate, R → Resultant = Reinforced

5 Types of Conflict

Conflict Type	Description	Example
Approach-Avoidance	Goal is both attractive and threatening	Attracted to someone but embarrassed
Avoidance-Avoidance	Two bad options	Climb high → fear, stay low → called coward
Approach-Approach	Two attractive options	Choosing between two favorite desserts

Tip: Remember **"Avoidance > Approach"** → Fear is stronger than desire.

6 Critical Stages of Development

- **First 6 years** are most important for personality formation.
- Mishandled **feeding, toilet training, sex education, aggression control** → neurotic conflicts.
- Early experiences shape **unconscious habits and anxieties**.

Memory Hook: "Early learning = lifelong patterns"

7 Role of Language & Social Context

- Language = key **cue-producing response**
- Culture shapes how children **label experiences** → guides behavior
- Parents may unintentionally create **neurotic conflicts** due to contradictory social rules.

Example:

- "Be aggressive when necessary but polite always" → confusion, frustration
-

8 Psychotherapy (S-R Perspective)

- Focus = **unlearn maladaptive habits** and form adaptive responses.
- Therapist = sympathetic, non-punitive, encourages free expression.
- Mechanism = understanding learned S-R connections.

Mnemonic: Therapy = Re-learning S-R

9 Mnemonics

1. **DCRR** → Drive → Cue → Response → Reward (Learning cycle)
2. **P-S-H** → Primary, Secondary, Habit (Motivation → Learned → Response)
3. **I-R** → Initial & Resultant hierarchy (Before vs After learning)
4. **A-A** → Approach–Avoidance conflict

5. **E1-E6** → Early critical stages = first 6 years

10 Examples Table

Concept	Example
Habit	Checking phone when notification rings
Primary Drive	Baby cries for food
Secondary Drive	Fear of dentist after a painful visit
Approach-Avoidance	Attraction + embarrassment
Avoidance-Avoidance	Climb high vs called coward
Initial Hierarchy	Baby cries at every noise
Resultant Hierarchy	Baby cries only at dangerous noise
Psychotherapy	Replacing exam anxiety with structured study habit